MITOCHONDRIAL & CELLULAR ENERGY

SS-31 — the full review

CARDIOLIPIN-TARGETING TETRAPEPTIDE (ELAMIPRETIDE)

This is the deeper look behind the short version — the approval history, including an earlier rejection, the two large trials that failed to hit their main goals, why the one approval that exists is a narrow, conditional one, and what your consultation covers.

What it is

SS-31, known by its drug development name elamipretide, is a four-amino-acid peptide engineered to be drawn specifically into the inner membrane of mitochondria, the structures inside cells that generate energy. It works differently from most compounds in this catalog: rather than acting on a hormone receptor, it binds a molecule called cardiolipin, found almost exclusively in that inner mitochondrial membrane, which organizes the machinery that produces cellular energy.

Under the brand name Forzinity, elamipretide is FDA-approved — the only compound in this catalog that can say that — but only for one ultra-rare, genetic, pediatric-onset disease called Barth syndrome, and only after an earlier rejection. What is sold through compounding pharmacies and peptide vendors as "SS-31" for general longevity or mitochondrial-health use is a different, untested product, aimed at a population and use the approved drug's trials never studied.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Barth syndrome — muscle strength	**FDA-approved**	Approved in 2025 through an accelerated pathway, based on a muscle-strength surrogate, after an earlier rejection; the required confirmatory trial is due in 2030 and approval could be withdrawn if it fails.
Age-related vision loss (geographic atrophy)	**Human trials**	The pivotal 176-person trial missed both of its main measures; a secondary anatomical finding is being tested in an ongoing, larger trial. Not an approved use.
Primary mitochondrial myopathy	**Human trials**	The pivotal 218-person trial failed both of its main measures; only an unreplicated subgroup showed benefit. Not an approved use.
Heart failure	**Human trials**	A small trial found no meaningful benefit; the use does not appear to be in active further development.
General mitochondrial health, longevity, or anti-aging	**Not studied**	No controlled human trial has tested this use in healthy adults; this is the population and goal the compounded product is actually sold for.

How it works

Cardiolipin folds the inner mitochondrial membrane into the folds that organize the electron transport chain, the machinery that produces ATP, the cell's energy currency. When cardiolipin is damaged — by oxidative stress, aging, or disease — that folded structure collapses, the electron transport chain leaks more cell-damaging byproducts, and that damages more cardiolipin, a self-reinforcing decline implicated in rare mitochondrial diseases and, more broadly, in general age- and disease-related mitochondrial dysfunction.

SS-31 binds cardiolipin directly, without disrupting the membrane itself, and is proposed to stabilize that folded structure, better organize the electron transport chain, reduce the leakage of damaging byproducts, and support ATP production. This mechanism is well characterized at the biochemical level — but as the research below shows, that has not reliably translated into positive results in the largest human trials that have tested it.

What the research actually looked at

The one approval that exists — for Barth syndrome, a rare genetic disease affecting heart and skeletal muscle — followed a difficult path: an original application was rejected by the FDA in 2025 after a lengthy review, and the drug was ultimately approved later that year only under an accelerated pathway, based on a muscle-strength measurement used as a surrogate for benefit, from a small 12-person trial whose blinded phase did not show a significant effect on its original primary measures. The confirmatory trial FDA required as a condition of approval only began in early 2026, with results not due until 2030, and the approval can be withdrawn if that trial fails.

In two other serious diseases, elamipretide's largest, most rigorous trials did not succeed. A 218-person Phase 3 trial in a genetic muscle disease called primary mitochondrial myopathy failed both of its main measures, with only an unreplicated subgroup finding a benefit. A separate 176-person Phase 2 trial in a form of age-related vision loss (geographic atrophy) also missed both of its main measures, though a secondary anatomical signal is being tested further in an ongoing, larger trial. A small heart-failure trial found no meaningful benefit, and elamipretide no longer appears in the company's stated development pipeline for that use.

None of this trial history — in any of these diseases — was conducted in healthy adults, or with a longevity or general-wellness goal. That evidence base, for the way SS-31 is actually marketed through compounding today, does not exist.

Common misunderstandings

- **“FDA-approved means broadly proven safe and effective.”** It is approved for one muscle-strength measurement in one ultra-rare pediatric genetic disease, through an accelerated pathway, after an earlier outright rejection — that says nothing about a healthy adult using it for a different goal.
- **“The mitochondrial mechanism is proven, so the clinical benefit must be too.”** The underlying biochemistry is well documented, but the largest controlled trials in two separate serious diseases — a muscle disease and an eye disease — both failed to meet their main goals.
- **“Compounded SS-31 is basically the same as the approved drug.”** The approved product is a specific formulation, at a specific dose, for a specific disease; compounded "SS-31" is a different, untested product, and compounding a near-copy of a drug that already has an approved version sits on uncertain legal footing.
- **“It's basically proven for wellness because it has so much trial history.”** It has more human trial data than almost anything else in this catalog, but that data comes from specific serious diseases, much of it endpoint-negative — not from the healthy-adult, longevity-focused use it is actually marketed for.

Who it is not for

- Anyone extrapolating the Barth syndrome approval, or any of the disease-specific trial results, to general wellness or longevity use — an untested population and goal.
- Pregnant or breastfeeding patients, or anyone trying to conceive — no human pregnancy or lactation data exists.
- Anyone with a known hypersensitivity — reactions including rash, facial swelling, and respiratory symptoms have been reported.
- Anyone treating the failed-endpoint trials (muscle disease, eye disease, heart failure) as evidence of proven benefit.
- Anyone sourcing product from an unverified compounder or vendor with no confirmed dose, purity, or sterility.
- Competitive athletes — status has changed recently with the new approval, so current WADA/USADA status should be independently verified.

Competitive athletes: SS-31 does not currently appear by name on the WADA Prohibited List, and now that it holds an FDA approval for Barth syndrome, it may no longer clearly fit the catch-all category applied to unapproved substances. This is a recent, still-settling status rather than a firm, permanent clearance — competitive athletes should verify current status with their sport's anti-doping authority before use.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

One narrow approval exists, reached only after an earlier rejection. Approved in 2025 for muscle strength in Barth syndrome specifically, through an accelerated pathway with a required confirmatory trial due in 2030 — not a clean, first-pass approval.

The largest trials in two other serious diseases both fell short. A 218-person muscle-disease trial and a 176-person eye-disease trial each failed to meet their main measures.

Real human trial history exists here, more than for most compounds in this catalog — but almost none of it is in the healthy-adult, longevity-focused population the compounded product is sold to.

Clinical detail

A four-amino-acid peptide (elamipretide, formerly Bendavia/MTP-131) that binds cardiolipin in the inner mitochondrial membrane.

FDA-approved as Forzinity (elamipretide hydrochloride) in September 2025 for muscle strength in Barth syndrome, in patients 30 kg and over, via accelerated approval, after an earlier Complete Response Letter in May 2025.

Confirmatory trial (4TAZPower) began dosing in early 2026; results due 2030, with approval revocable if the trial fails.

Phase 3 trial in primary mitochondrial myopathy (n=218) failed both co-primary endpoints; Phase 2 trial in geographic atrophy (n=176) failed both co-primary endpoints, with a secondary signal under further study.

Not on FDA's 503A or 503B compounding bulk-substance lists or the July 2026 PCAC review docket; compounding a near-copy of an already-approved drug carries its own distinct legal restriction.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

SS-31 is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.